

stepper motors are connected to the Pololu stepper control. If you are assembling your own board, you can pick them up for about \$5 a pop.

The End Stops

The end stops are the places in the print area where the X,Y and Z coordinates are (0,0,0). On the X axis it is at the left most extreme, Y is as far back as the print plate can go and Z is almost touching the print plate. In fact the Z axis jumps up a little bit before beginning to print.

One end stop per axis is enough and the firmware installed in the micro-processor will automatically know how far each axis is allowed to move from zero.

The end stops could either be mechanical or optical end stops but the Reprap community seems to have a leaning towards mechanical ones.

Filaments

After the extruder, arguably the most important part in any 3D printer is the filament. ABS (Acrylonitrile Butadiene Styrene) plastic and PLA (Polylactic Acid) are the two most common types of plastics available and each has its own advantages.

While ABS plastic is the more durable of the two, PLA is a low more flexible (ABS is actually quite rigid) and is biodegradable.

The only flipside to PLA is that it responds poorly to heat and meets the same fate as the Wicked Witch of the West when mixed with water, when the thermometer goes north of 50 degrees Celsius.

The flipside to ABS is that because it is so rigid, it's more prone to warping so a wise choice we would guess would be to invest in a heated base.

The key to good quality printing is really in the balance between the rate at which the plastics are extruded, the temperature of the hot-end and the speeds at which the extruder moves across the axes so try to play around with the variables. Start from a slow speed and work your way up.

The Filabot is a system that allows the user to make the filament at home and according to them, save up to 90% on filament costs. It's pretty flexible too and produces filaments of ABS and PLA of varying thicknesses. You can



Available in more colours and opacities than you can imagine